

C960

Discrete Mathematics II

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A team tested a routing algorithm on 50 inputs using a cloud cluster that costs \$3 an hour. Its running time is $T(n) = 5n^2 + 200n \log n + 1000$. What is the tightest big-O estimate?

- A. $O(1)$
- B. $O(n^2)$
- C. $O(n^3)$
- D. $O(n \log n)$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A contractor who has finished 60 kitchens this year must cover a 252 cm by 198 cm floor with the largest identical square tiles possible, without cutting any. Tiles cost \$4 each. What is the tile side length?

- A. 6 cm
- B. 54 cm
- C. 18 cm
- D. 2,772 cm

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

In a training lab at a university with 18,000 students, keys generated on a Tuesday use RSA with $p = 5$, $q = 11$, and public exponent $e = 3$. What is the private key d ?

- A. 13
- B. 37
- C. 27
- D. 55

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

An RSA modulus is $n = 77$, built from primes 7 and 11. What is $\phi(n)$?

- A.** 60
- B.** 18
- C.** 77
- D.** 76

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A tutor at a community college with 9,000 students wants to prove $1 + 2 + \dots + n = n(n + 1)/2$ for all $n \geq 1$. She has already checked $n = 1$ through 10 by computer and the base case holds. What must the inductive step show?

- A.** Nothing more; ten checked cases prove it.
- B.** Assuming it holds for $n = k + 1$, it holds for $n = k$.
- C.** Assuming the formula holds for $n = k$, it holds for $n = k + 1$.
- D.** Assume it holds for all n , then simplify.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A nonprofit founded in 1998 holds board meetings on Thursdays. It must pick a 3-person review committee from 10 volunteers; members have no special roles. How many committees are possible?

- A.** 120
- B.** 1,000
- C.** 720
- D.** 360

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

A club found 720 ways to fill the roles of president, vice president, and treasurer from its members, with no one holding two roles. How many members does it have?

- A.** 9
- B.** 6
- C.** 720
- D.** 10

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A firm headquartered in Denver, where average tenure is six years, has 50 employees. What is the largest number that is guaranteed to share the same birth month?

- A. 5
- B. 4
- C. 13
- D. 4.17

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

What is the smallest number of people that guarantees at least 3 share a birth month?

- A. 13
- B. 36
- C. 24
- D. 25

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A mail server holding 2 TB of email uses 3 disks bought from an overseas vendor. Each disk independently has a 0.1 chance of failing this year. What is the probability that at least one fails?

- A. 0.001
- B. 0.271
- C. 0.729
- D. 0.3

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

Two independent tries each have the same success chance p . The chance of at least one success is 0.75. What is p ?

- A. About 0.866
- B. 0.5
- C. 0.375
- D. 0.25

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A vending controller designed by an engineering firm in Ohio is a DFA over $\{0, 1\}$ that accepts exactly the strings with an even number of 1s. Which string does it accept?

- A.** 010
- B.** None containing 0s, since the machine only counts 1s
- C.** 0111
- D.** 0110

Target word	The one fact that decides	Answer	☐ sure ☐ unsure ☐ guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

A team ~~tested a routing algorithm on 50 inputs~~ using a cloud cluster ~~that costs \$3 an hour~~. Its running time is $T(n) = 5n^2 + 200n \log n + 1000$. What is the tightest big-O estimate?

Target word: tightest big-O **Decides:** n^2 grows fastest.

A	Wrong stem number	Focuses on the constant 1000.
B	Correct	Coefficients and slower terms drop out.
C	Adjacent category	A valid upper bound, but not the tightest.
D	Confusion set	Picks the term with the largest coefficient.

2. Answer C Concept 2

A contractor ~~who has finished 60 kitchens this year~~ must cover a 252 cm by 198 cm floor with the largest identical square tiles possible, without cutting any. ~~Tiles cost \$4 each~~. What is the tile side length?

Target word: gcd(252, 198) **Decides:** $252 \rightarrow 198 \rightarrow 54 \rightarrow 36 \rightarrow 18 \rightarrow 0$.

A	Confusion set	A common divisor, but not the greatest.
B	Dropped element	Stops after the first remainder.
C	Correct	The last nonzero remainder is 18.
D	Inverted operation	That is the lcm.

3. Answer C Concept 3

In a training lab ~~at a university with 18,000 students~~, keys ~~generated on a Tuesday~~ use RSA with $p = 5$, $q = 11$, and public exponent $e = 3$. What is the private key d ?

Target word: d **Decides:** $\phi = 4 \times 10 = 40$; $3d \equiv 1 \pmod{40}$.

A	Inverted operation	$3 \times 13 = 39 \equiv -1$, not 1.
B	Confusion set	Solves mod $n = 55$ instead of mod $\phi(n)$.
C	Correct	$3 \times 27 = 81 = 2 \times 40 + 1$.
D	Wrong stem number	That is n .

4. Answer A Concept 3 • transfer

An RSA modulus is $n = 77$, built from primes 7 and 11. What is $\phi(n)$?

Target word: $\phi(n)$ **Decides:** $(7 - 1)(11 - 1)$.

A	Correct	$6 \times 10 = 60$.
B	Adjacent category	$p + q$ is not the totient.
C	Wrong stem number	That is n .
D	Confusion set	$n - 1$ applies only when n is prime.

5. Answer C Concept 4

A tutor ~~at a community college with 9,000 students~~ wants to prove $1 + 2 + \dots + n = n(n + 1)/2$ for all $n \geq 1$. She has already checked ~~$n = 1$ through 10 by computer~~ and the base case holds. What must the inductive step show?

Target word: inductive step **Decides:** Assume $P(k)$, prove $P(k + 1)$.

A	Camouflage	Examples never prove a statement for all n .
B	Inverted operation	Runs the implication backward.
C	Correct	That links each case to the next.
D	Confusion set	Assumes the conclusion.

6. Answer A Concept 5

A nonprofit ~~founded in 1998 holds board meetings on Thursdays~~. It must pick a 3-person review committee from 10 volunteers; members have no special roles. How many committees are possible?

Target word: $C(10, 3)$ **Decides:** No roles: order does not matter.

A	Correct	$10 \times 9 \times 8 \div 3! = 120$.
B	Adjacent category	10^3 allows the same person more than once.
C	Confusion set	Permutations; treats order as mattering.
D	Dropped element	Divides by 2 instead of 3!.

7. Answer D Concept 5 · transfer

A club found 720 ways to fill the roles of president, vice president, and treasurer from its members, with no one holding two roles. How many members does it have?

Target word: solve for n **Decides:** $n(n - 1)(n - 2) = 720$.

A	Adjacent category	Treats it as $n^3 \approx 720$, allowing repeats.
B	Confusion set	$3!$ is the number of role orders.
C	Wrong stem number	The count of ways.
D	Correct	$10 \times 9 \times 8 = 720$.

8. Answer A Concept 6

A firm ~~headquartered in Denver, where average tenure is six years~~, has 50 employees. What is the largest number that is guaranteed to share the same birth month?

Target word: $\lceil 50/12 \rceil$ **Decides:** 50 items into 12 boxes.

A	Correct	$\lceil 50 \div 12 \rceil = \lceil 4.17 \rceil = 5$.
B	Inverted operation	Rounds down instead of up.
C	Adjacent category	People needed to force a pair, not the guaranteed group size.
D	Dropped element	People come in whole numbers; the ceiling is required.

9. Answer D Concept 6 · transfer

What is the smallest number of people that guarantees at least 3 share a birth month?

Target word: solve for N **Decides:** $12 \times (3 - 1) + 1$.

A	Adjacent category	Guarantees a pair, not three.
B	Inverted operation	Multiplies 12×3 .
C	Dropped element	Forgets the +1.
D	Correct	24 people could have 2 per month; the 25th forces a third.

10. Answer B Concept 7

A mail server ~~holding 2 TB of email~~ uses 3 disks ~~bought from an overseas vendor~~. Each disk independently has a 0.1 chance of failing this year. What is the probability that at least one fails?

Target word: P(at least one) **Decides:** $1 - 0.9^3$.

A	Confusion set	Probability all three fail.
B	Correct	$1 - 0.729 = 0.271$.
C	Inverted operation	Probability none fail.
D	Adjacent category	Adds the probabilities, double-counting overlaps.

11. Answer B Concept 7 · transfer

Two independent tries each have the same success chance p . The chance of at least one success is 0.75. What is p ?

Target word: solve for p **Decides:** $(1 - p)^2 = 0.25$.

A	Inverted operation	Takes $\sqrt{0.75}$ instead of working through the complement.
B	Correct	$1 - p = 0.5$.
C	Adjacent category	Splits 0.75 across two tries.
D	Confusion set	That is $(1 - p)^2$, the chance of no success.

12. Answer D Concept 8

A vending controller ~~designed by an engineering firm in Ohio~~ is a DFA over $\{0, 1\}$ that accepts exactly the strings with an even number of 1s. Which string does it accept?

Target word: accepted string **Decides:** Count the 1s.

A	Adjacent category	Ends in 0, but has one 1.
B	Absolute language	0s simply keep the current state.
C	Confusion set	Even length, but three 1s.
D	Correct	Two 1s: even.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	4	
Wrong stem number	A figure copied from the prompt that is not the answer.	4	
Adjacent category	The right idea in the wrong bucket.	9	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	7	
Absolute language	Always, never, guarantee — where judgment applies.	1	
Confusion set	Every option is real; the test is exact role-matching.	10	
Camouflage	Answers a stem fact that was never the question.	1	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Big-O estimates	The tightest big-O keeps only the fastest-growing term and drops constants and coefficients.	Picking the term with the biggest coefficient.	A running-time formula with several terms.
2	GCD by the Euclidean algorithm	$\gcd(a, b) = \gcd(b, a \bmod b)$, repeated until the remainder is 0; the last nonzero remainder is the gcd.	Stopping after one step, or confusing gcd with lcm.	“Largest equal pieces” or “largest square tile.”
3	RSA key math	$n = pq$, $\phi(n) = (p - 1)(q - 1)$, and the private key d satisfies $e \cdot d \equiv 1 \pmod{\phi(n)}$.	Working mod n instead of mod $\phi(n)$.	Two small primes and a public exponent.
4	Mathematical induction	Prove a base case, then assume $P(k)$ and prove $P(k + 1)$.	Treating many checked examples as a proof, or assuming what is to be proved.	“For all integers $n \geq 1$.”
5	Permutations vs. combinations	Order matters $\rightarrow P(n, r) = n!/(n - r)!$; order doesn’t matter $\rightarrow C(n, r) = n!/(r!(n - r)!)$.	Using permutations for a committee with no roles.	Distinct roles vs. an unordered group.
6	Pigeonhole principle	If N items go into k boxes, some box holds at least $\lceil N/k \rceil$; to force r in one box, you need $k(r - 1) + 1$ items.	Rounding down or forgetting the $+1$.	“Guaranteed” or “at least” with categories like months.
7	Complement rule for “at least one”	$P(\text{at least one}) = 1 - P(\text{none})$; for independent events multiply the “none” probabilities.	Adding the individual probabilities.	“At least one” across independent trials.
8	Finite-state automata	A DFA accepts a string if the input ends in an accepting state; the machine tracks one property, such as parity.	Judging by string length or last symbol instead of the tracked property.	A machine description and candidate strings.

Course scope (public description): Discrete Mathematics II addresses abstract, discrete, computational methods used in computer science. In particular, this class introduces searching and sorting algorithms; big-O estimates; number theory and cryptography; recursion and induction; counting and advanced counting techniques; discrete probability; and modeling computation. This course emphasizes applications in computer science. Discrete Mathematics I is a prerequisite for this course.